

Control Flow

- Conditional Statements:
 - If Statements
 - Conditional Expressions
- Loops:
 - While Loops
 - For Loops
 - Nested Loops
- Match-Case Statements

Conditional Statements

- If Statements

- If = Do some code only IF some condition is True
Else do something else

Exercises

- Python Calculator
- Write a program that suggests what to wear based on whether it is raining and the temperature, providing appropriate clothing choices like a raincoat, sweater, or light clothes.
- Write a program that checks a user's age to determine their eligibility for signing up, with different messages based on whether they are too young, too old, or eligible.

Conditional Expressions

- One-line shortcut for the if-else statement
- Print or assign one of two values based on a condition
- X if condition else Y

Loops

- While Loop
 - Execute some code WHILE some condition remains true

Exercise

- Python compound interest calculator

Loops

- For Loop

- Execute a block of code a fixed number of times.
- Iterate over a range, string, sequence, etc.

Exercise

- Countdown

Nested Loops

A loop within another loop (outer, inner)

- while x > 0:
 - while y > 0:
 - print("do something")
- for x in range(3):
 - for y in range(9):
 - print("do something")

Match-Case Statements (Switch)

- An alternative to using many 'elif' statements
- Execute some code if a value matches a 'case'
- Cleaner and syntax is more readable